

Domain Adaptation and Generalization Using Foundation Models in Healthcare Imaging

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Abstract

The application of artificial intelligence (AI) in medical imaging has seen remarkable advances in recent years, particularly through the use of deep learning models trained on large-scale datasets. However, despite impressive performance on in-distribution data, such models often struggle to maintain reliability and accuracy when deployed in new clinical environments—a phenomenon commonly attributed to domain shift. Domain adaptation (DA) and domain generalization (DG) have thus emerged as critical areas of research, aiming to improve model robustness across diverse imaging modalities, scanners, patient populations, and institutions. At the same time, the rise of foundation models—large-scale, pre-trained architectures capable of zero-shot or few-shot learning across tasks and domains—has opened up new possibilities for tackling these challenges with unprecedented flexibility and scale. This survey provides an in-depth exploration of the intersection between DA/DG methods and foundation models in medical imaging. We begin by outlining the theoretical motivations for domain-robust learning and describe the unique characteristics of medical imaging that intensify the need for generalization, such as high inter-institutional variability, rare disease prevalence, and modality-specific artifacts. We then review foundational techniques for DA and DG, including supervised and unsupervised strategies, self-supervised learning, adversarial alignment, and meta-learning approaches. Building on this foundation, we analyze how recent advancements in foundation models—particularly vision transformers, contrastive pretraining, and vision-language models—are reshaping the landscape of domain-robust AI in medicine. Key themes include the reuse of pre-trained representations, adapter-based fine-tuning, and prompt engineering for zero-shot generalization. Beyond algorithmic strategies, the survey critically examines the limitations and ethical concerns that accompany the deployment of foundation models in clinical settings. These include data bias, interpretability challenges, prompt sensitivity, infrastructural barriers, and legal ambiguity. We highlight that while foundation models offer

improved adaptability, they are not a panacea; rather, their successful deployment demands careful attention to fairness, transparency, and continuous evaluation. Furthermore, we identify a set of open research questions and future directions—such as causal reasoning for generalization, domain-aware pretraining, interactive model adaptation, and participatory development frameworks—that will be pivotal in closing the gap between research prototypes and clinically robust AI systems. Through this comprehensive analysis, we aim to provide researchers, clinicians, and developers with a conceptual roadmap for building and evaluating generalizable AI systems in medical imaging. By situating DA and DG within the broader context of foundation model development, this survey underscores the importance of interdisciplinary collaboration and ethically grounded innovation in shaping the future of medical AI.

Keywords: Domain Adaptation, Domain Generalization, Foundation Models, Medical Imaging, Transfer Learning, Self-Supervised Learning, Vision Transformers, Multimodal Learning, Representation Learning, Clinical AI Robustness, Zero-Shot Learning, Cross-Domain Generalization, Model Generalizability, Radiology AI, Healthcare Machine Learning

1 Introduction

Medical imaging plays a pivotal role in modern healthcare, providing critical insights into the diagnosis, treatment planning, and monitoring of a wide array of diseases and conditions. With the growing availability of medical image data and the increasing power of deep learning algorithms, there has been a surge of interest in leveraging artificial intelligence (AI) to automate and enhance image interpretation [1]. However, the deployment of AI models in real-world clinical settings continues to face significant challenges, particularly due to variations across imaging domains. These variations may arise from differences in scanner types, imaging protocols, patient populations, disease prevalence, and annotation quality across institutions and regions [2]. Such heterogeneity leads to domain shifts that can severely degrade the performance of AI models when applied outside their training distribution. To address this challenge, research in domain adaptation (DA) and domain generalization (DG) has emerged as a cornerstone of robust medical image analysis. Domain adaptation techniques aim to align or adapt a model trained on a source domain to perform well on a target domain, typically with limited or no labeled data in the target [3]. In contrast, domain generalization focuses on developing models that inherently generalize to unseen domains without requiring access to target domain data during training. These strategies are essential for ensuring the safe, effective, and equitable deployment of AI systems across di-

verse healthcare environments, particularly when retraining or fine-tuning models for every new clinical site is impractical or impossible [4]. In recent years, the emergence of foundation models—large-scale models trained on vast and diverse datasets with self-supervised, supervised, or multimodal learning objectives—has brought a transformative shift to many fields, including natural language processing, computer vision, and increasingly, medical imaging [5]. Foundation models such as CLIP, SAM, BioViL, and MedCLIP have demonstrated remarkable capabilities in learning transferable representations that can generalize across tasks, modalities, and domains. Their potential to serve as general-purpose backbones for downstream applications in medical imaging is currently being actively explored. The intersection of foundation models with domain adaptation and generalization in medical imaging represents a highly promising but still nascent research frontier. On one hand, foundation models can be leveraged as pretrained feature extractors whose representations are more robust to domain shifts [6]. On the other hand, these models can themselves be adapted using domain-specific fine-tuning, prompt learning, or contrastive learning strategies to further improve performance across varying data distributions. Moreover, the modularity and scalability of foundation models open new avenues for hybrid approaches that integrate domain adaptation modules or use synthetic data generation to support generalization. Despite their promise, the use of foundation models in domain adaptation and generalization for medical imaging also raises several open questions and challenges. These include understanding the limits of zero-shot transfer, quantifying domain shift and its impact on foundation model representations, designing effective adaptation pipelines that respect clinical constraints, and ensuring fairness and transparency when models are deployed across diverse patient populations and healthcare systems [7]. Furthermore, the data requirements, computational costs, and ethical considerations associated with training and deploying large foundation models need careful examination in the medical context [8]. This survey aims to provide a comprehensive overview of the current landscape of domain adaptation and generalization in medical imaging, with a particular focus on the role of foundation models. We begin by reviewing the theoretical foundations of domain shift, adaptation, and generalization, including key concepts such as covariate shift, label shift, and representation alignment. We then examine classical and contemporary methods for domain adaptation and generalization, including feature-based, pixel-based, and model-based approaches. Next, we delve into the design and use of foundation models in medical imaging, exploring how these models are pre-trained, fine-tuned, and evaluated [9]. We synthesize the growing body of work that employs foundation models for robust and generalizable medical image analysis across domains, and highlight both the successes and limitations observed to date. In addition to technical developments, we discuss practical considerations

in building and deploying generalizable models in clinical environments, including challenges in dataset curation, annotation variability, regulatory requirements, and model interpretability [10]. We also explore the implications of foundation models for clinical practice, such as the potential for enabling cross-site federated learning, supporting underserved regions with limited data, and facilitating more adaptive and responsive AI systems in dynamic clinical workflows. Finally, we identify emerging trends and future directions for research in this area. These include the development of universal medical vision-language models, integration with electronic health records (EHRs) and multimodal data sources, advancements in unsupervised domain generalization, and the potential of generative models such as diffusion models and generative adversarial networks (GANs) for domain simulation and augmentation. We aim to provide a forward-looking perspective on how the convergence of domain adaptation, generalization, and foundation models may shape the future of AI in medical imaging—enabling more robust, equitable, and trustworthy systems that can be deployed at scale across diverse clinical settings. In sum, this survey seeks to map out the evolving terrain of domain adaptation and generalization in medical imaging, contextualized within the broader shift toward foundation model paradigms. By systematically reviewing the literature, highlighting methodological innovations, and articulating open research questions, we hope to provide a valuable resource for researchers, clinicians, and developers working at the interface of medical AI and real-world deployment [11].

2 Theoretical Background

The study of domain adaptation and generalization is rooted in statistical learning theory, where the central challenge lies in ensuring that a model trained on one data distribution performs reliably on another [12]. Let us denote by $\mathcal{X}$ the input space (e.g., medical images such as CT, MRI, or ultrasound scans) and by $\mathcal{Y}$ the output space (e.g., class labels for disease detection or segmentation masks) [13]. A domain is typically defined as a pair $\mathcal{D} = (\mathcal{P}_X, f)$, where $\mathcal{P}_X$ is a probability distribution over $\mathcal{X}$, and $f : \mathcal{X} \rightarrow \mathcal{Y}$ is the labeling function, which may be unknown in practice [14]. In the case of supervised learning, we assume access to a labeled dataset $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ sampled from a source domain $\mathcal{D}_S = (\mathcal{P}_S, f_S)$. The goal of domain adaptation is to leverage $\mathcal{S}$ to learn a hypothesis $h \in \mathcal{H}$ that minimizes the expected risk on a different, but related, target domain $\mathcal{D}_T = (\mathcal{P}_T, f_T)$, for which we may have limited or no labeled samples. The target risk is defined as:

$$\mathcal{R}_T(h) = \mathbb{E}_{x \sim \mathcal{P}_T}[\ell(h(x), f_T(x))],$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ is a loss function (e.g., cross-entropy or Dice loss). Since $\mathcal{R}_T(h)$ cannot be directly minimized without access to $(x, f_T(x))$ pairs, domain

adaptation methods aim to minimize a proxy objective that incorporates the empirical source risk $\mathcal{R}_S(h)$ and a divergence measure between the source and target distributions [15]. One of the foundational results in domain adaptation is the generalization bound by Ben-David et al., which relates the target risk to the source risk and the domain discrepancy:

$$\mathcal{R}_T(h) \leq \mathcal{R}_S(h) + \mathcal{D}_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{P}_S, \mathcal{P}_T) + \lambda,$$

where $\mathcal{D}_{\mathcal{H}\Delta\mathcal{H}}$ denotes the $\mathcal{H}\Delta\mathcal{H}$ divergence between domains, and λ quantifies the difference in labeling functions across domains [16]. This bound provides a theoretical justification for aligning feature distributions across domains while maintaining discriminative power on the source [17]. In medical imaging, the shift between $\mathcal{P}_S$ and $\mathcal{P}_T$ may arise from various factors such as different scanners, acquisition protocols, anatomical variability, demographic distribution, or pathology prevalence. These shifts can manifest as covariate shift (i.e., $\mathcal{P}_S(x) \neq \mathcal{P}_T(x)$ but $f_S = f_T$), label shift ($\mathcal{P}_S(y) \neq \mathcal{P}_T(y)$), or conditional shift ($\mathcal{P}_S(y | x) \neq \mathcal{P}_T(y | x)$) [18]. Each type of shift necessitates different adaptation strategies, ranging from feature alignment and sample reweighting to invariant representation learning and adversarial training. Domain generalization extends these ideas by aiming to learn a hypothesis h that generalizes well to any domain drawn from a distribution over domains $\mathcal{Q}$, without access to the target domain at training time. Formally, the objective becomes:

$$\min_{h \in \mathcal{H}} \mathbb{E}_{\mathcal{D} \sim \mathcal{Q}}[\mathcal{R}_{\mathcal{D}}(h)],$$

where the expectation is over domains $\mathcal{D} = (\mathcal{P}, f)$ not seen during training. Since the target domain is unknown, DG methods aim to learn features that are domain-invariant or domain-agnostic, often through meta-learning, data augmentation, contrastive learning, or information-theoretic regularization. A critical aspect of domain generalization is the trade-off between domain invariance and task discriminability: overly invariant representations may suppress medically relevant variations, leading to degraded performance [19]. Foundation models introduce a new paradigm in this context [20]. Trained on massive datasets $\mathcal{S}_{\text{pretrain}} = \{(x_i^{(k)}, y_i^{(k)})\}_{i,k}$ covering multiple imaging modalities, anatomical regions, and tasks, these models aim to learn general-purpose embeddings $z = \phi(x)$ that can transfer across domains and tasks. The learned representation function $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ often serves as the backbone for downstream applications. Given the inductive bias and scale of these models, it is hypothesized that ϕ maps instances from different domains into a shared latent space where domain discrepancy is minimized:

$$\|\mathbb{E}_{x \sim \mathcal{P}_S}[\phi(x)] - \mathbb{E}_{x \sim \mathcal{P}_T}[\phi(x)]\|_2 \approx 0.$$

This alignment suggests that adaptation may require only lightweight fine-tuning of task-specific heads $h_{\text{task}} \circ \phi$, or in some cases, prompt-based adaptation or zero-

shot inference. Nonetheless, understanding when and why foundation models succeed in domain adaptation and generalization remains an open theoretical question. Empirical evidence suggests that pretraining objectives, such as contrastive learning or masked image modeling, implicitly encourage alignment across domains by focusing on semantic consistency rather than pixel-level features. Furthermore, multimodal models trained with paired image-text data (e.g., radiology reports and imaging studies) may develop cross-domain invariance through language grounding, where textual descriptions act as regularizers or auxiliary supervision [21]. However, the robustness of these representations under severe domain shifts—such as transitioning from adult to pediatric populations, or from high-resource to low-resource settings—must be rigorously quantified [22]. In this section, we have formalized the problems of domain adaptation and generalization using mathematical notation and introduced the theoretical underpinnings that motivate algorithmic solutions. In the next sections, we will examine in detail the methodological landscape, including traditional approaches and recent innovations leveraging foundation models to tackle domain shifts in medical imaging.

3 Methodologies for Domain Adaptation and Generalization in Medical Imaging

The increasing recognition of domain shift in medical imaging has led to the development of a broad spectrum of methodologies aimed at either adapting models to new domains (domain adaptation) or building models that inherently generalize to unseen domains (domain generalization) [23]. These methods can be broadly categorized into feature-based approaches, pixel-level strategies, model-centric designs, and more recently, foundation model-based paradigms [24]. In this section, we provide an extensive review of these methodologies, illustrating how they are used to enhance model robustness across domains in medical imaging tasks such as classification, segmentation, detection, and retrieval.

3.1 Feature-Level Alignment Approaches

Feature-based methods seek to learn domain-invariant representations such that a model trained on source domain features can generalize to target domain samples. A common approach involves minimizing the distance between source and target feature distributions, typically using discrepancy metrics such as Maximum Mean Discrepancy (MMD), Correlation Alignment (CORAL), or adversarial losses. In adversarial domain adaptation, inspired by the success of Generative Adversarial Networks (GANs), a domain discriminator is trained to distinguish between

source and target features, while the feature encoder learns to fool the discriminator, effectively aligning the domains in the latent space. In medical imaging, these strategies have been applied to tasks like cross-modality segmentation (e.g., MRI to CT), where anatomical structures remain consistent but image intensities vary. For example, domain adversarial neural networks (DANNs) have been used to extract domain-agnostic features for liver or prostate segmentation, improving generalization without labeled data in the target modality [25]. More recent variants, such as conditional domain adversarial networks (CDANs) and domain-specific batch normalization, extend these ideas by conditioning the alignment on class predictions or learning domain-aware normalization statistics [26].

3.2 Pixel-Level Adaptation and Style Transfer

Pixel-level adaptation operates in the input space, transforming source domain images to resemble those from the target domain using image-to-image translation techniques [27]. This is particularly useful in medical imaging, where differences in scanner calibration, imaging protocols, or noise artifacts can create large covariate shifts [28]. CycleGAN and its variants have been widely used to generate synthetic target-like images while preserving anatomical content. These synthetic images can then be used to train or fine-tune models, reducing the domain gap. For instance, CT-to-MRI or T1-to-T2 translation has been explored to improve cross-modality segmentation performance. However, the clinical reliability of such synthetic images remains a concern, as image translation may introduce artifacts or hallucinate pathological features. To mitigate this, hybrid approaches combine pixel-level and feature-level adaptation, ensuring both visual realism and semantic consistency [29].

3.3 Model-Based and Regularization Approaches

Model-centric methods focus on making the learning process itself more robust to domain shifts. This includes data augmentation strategies, domain-specific normalization, dropout regularization, and meta-learning frameworks. Data augmentation plays a particularly important role in medical imaging, where access to diverse datasets may be limited. Techniques such as elastic deformation, intensity scaling, and domain-specific augmentations (e.g., simulating artifacts or varying resolution) are used to synthetically increase domain diversity. Meta-learning approaches, such as Model-Agnostic Meta-Learning (MAML), simulate domain shifts during training to learn models that can quickly adapt to new domains. These methods have shown promise in applications like few-shot lesion detection and rare disease classification. Another line of work involves self-supervised pretraining with tasks like rotation prediction, context restoration, or contrastive instance

discrimination, which help models learn more robust, transferable features [30].

3.4 Domain Generalization via Invariant Risk Minimization

Domain generalization techniques often draw on the principle of Invariant Risk Minimization (IRM), which aims to identify representations that lead to consistent performance across all training domains. In this setting, the training set is assumed to be composed of multiple domains $\{\mathcal{D}_i\}_{i=1}^K$, and the objective is to learn a feature extractor ϕ and classifier h such that $h \circ \phi$ achieves low risk on each domain, while preserving invariance across domains. Approaches such as Domain-Invariant Component Analysis (DICA), adversarial feature disentanglement, and group distributionally robust optimization (GroupDRO) have been used to enforce this consistency. These methods are especially relevant in federated medical imaging, where data from different hospitals or centers (i.e., different domains) cannot be centralized but must still contribute to a global model.

3.5 Foundation Model-Based Strategies

The advent of foundation models has introduced a new class of methodologies that exploit large-scale pretraining on diverse datasets to achieve strong zero-shot or few-shot generalization [31]. These models are trained with objectives such as contrastive learning (e.g., CLIP), masked image modeling (e.g., MAE, SimMIM), or multimodal supervision (e.g., paired images and radiology reports). Once pre-trained, foundation models can be adapted to downstream tasks with minimal supervision via prompt tuning, adapter modules, or lightweight fine-tuning. In the context of domain adaptation, foundation models are often used as fixed feature extractors, under the assumption that their representations are sufficiently robust to domain shift. Empirical results show that models such as MedCLIP or BioViL can outperform fully supervised baselines in cross-institutional settings without requiring target domain labels. In addition, multimodal models can be leveraged for zero-shot classification using text prompts that describe disease categories or imaging findings, offering a powerful alternative to task-specific retraining. Several recent works have also explored fine-tuning foundation models using domain-specific contrastive objectives, such as aligning pathological patterns in histopathology or adapting to low-resource modalities (e.g., PET or ultrasound). Another promising direction involves hybrid pipelines where a foundation model provides general representations that are then refined through domain-adaptive heads or auxiliary loss functions designed to detect domain shifts explicitly.

3.6 Combining Domain Adaptation and Generalization with Foundation Models

Increasingly, researchers are combining domain adaptation and generalization techniques with foundation models in modular architectures. For example, a foundation model may serve as a frozen encoder, while task-specific adapters are trained using domain generalization strategies such as meta-learning or invariant risk minimization [32]. Alternatively, intermediate layers of the foundation model may be fine-tuned using adversarial domain alignment to improve cross-institutional segmentation [33]. In scenarios where target domain data is completely unavailable, test-time adaptation or self-training methods are employed, leveraging the stability of foundation model embeddings and the ability to generate pseudo-labels or uncertainty estimates. Furthermore, synthetic data generation using diffusion models or GANs, guided by foundation model representations, is being explored as a way to simulate domain shifts during training, thereby enhancing generalization.

3.7 Evaluation Protocols and Benchmarks

Robust evaluation of DA and DG methods in medical imaging remains a challenge due to the lack of standardized benchmarks and the high variability across tasks [34]. Common practices include leave-one-domain-out validation (e.g., training on multiple hospitals and testing on an unseen one), synthetic-to-real transfer (e.g., using simulated phantoms or augmented data), and cross-modality generalization (e.g., CT to MRI) [35]. Key metrics include accuracy, Dice score, AUROC, and domain sensitivity analysis, often supplemented with uncertainty quantification [36]. The use of foundation models necessitates additional considerations, such as evaluating zero-shot performance, prompt sensitivity, calibration, and model interpretability across domains [37]. As the field matures, there is a growing need for community-driven benchmarks and datasets explicitly designed to evaluate domain adaptation and generalization under clinically realistic conditions [38].

In summary, the methodological landscape for domain adaptation and generalization in medical imaging has evolved significantly, from handcrafted alignment strategies to data-driven and representation-centric paradigms powered by foundation models [39]. These approaches not only mitigate the limitations of traditional supervised learning but also open the door to scalable, deployable, and more equitable AI systems across diverse medical environments [40]. In the following sections, we will explore concrete applications and case studies where these methodologies have been applied, along with empirical insights and practical lessons learned [41].

4 Applications and Case Studies

The theoretical constructs and methodological advances in domain adaptation (DA) and domain generalization (DG) have found tangible application across a wide array of medical imaging tasks. In this section, we delve into a series of representative applications and case studies that exemplify how foundation models—trained at scale and designed for general-purpose visual understanding—are being employed, evaluated, and adapted within clinical and biomedical imaging contexts [42]. These cases underscore both the opportunities and challenges associated with the deployment of adaptable AI systems across diverse healthcare settings.

4.1 Cross-Institutional Lesion Detection and Classification

Lesion detection and classification constitute one of the most common applications of machine learning in radiology, particularly in modalities such as computed tomography (CT), magnetic resonance imaging (MRI), and mammography. However, variations in patient demographics, acquisition settings, and annotation quality across institutions often lead to degraded performance when a model trained on data from one hospital is applied elsewhere [43]. To address this, foundation models pretrained on large-scale, multi-institutional datasets (e.g., TCIA, RSNA, and MIMIC-CXR) have been used as feature extractors or initializations for lesion classification models [44]. In one notable case study, a CLIP-style model trained on over 1 million image-report pairs from multiple institutions was evaluated on out-of-distribution (OOD) datasets for lung nodule detection [45]. The model achieved strong zero-shot and few-shot performance, maintaining over 90% of its in-domain accuracy despite a substantial shift in image acquisition protocol and population health profiles. Further improvements were achieved via prompt tuning using textual descriptions of nodule appearance from radiology guidelines, demonstrating the utility of cross-modal representations for generalization.

4.2 Cross-Modality Segmentation with Unpaired Data

Another prominent area of application is anatomical segmentation across imaging modalities, such as transferring models trained on MRI to CT or vice versa. These scenarios are inherently challenging due to differences in intensity distributions, spatial resolution, and modality-specific artifacts. Traditional approaches relying on pixel intensity or handcrafted features fail to generalize across such shifts. Foundation models pretrained using masked autoencoders (MAE) or contrastive learning objectives (e.g., SimCLR, MoCo) have shown remarkable resilience to modality changes when used as frozen backbones for segmentation tasks. For ex-

ample, a MAE-based model pretrained on a large unannotated corpus of CT and MRI scans was fine-tuned on brain MRI segmentation and evaluated on CT without any additional CT supervision. Despite the lack of target domain labels, the model achieved Dice scores within 5–10% of supervised baselines, outperforming adversarial domain adaptation techniques. This illustrates how semantic priors encoded during self-supervised pretraining can bridge cross-modality gaps.

4.3 Pediatric-to-Adult Transfer and Rare Disease Settings

One of the most impactful use cases of domain generalization lies in extending AI capabilities to underrepresented populations, including pediatric imaging and rare disease detection. Data scarcity in these settings precludes extensive supervised training and exacerbates domain shifts due to anatomical or pathological differences [46]. Recent studies have used foundation models pretrained on adult imaging datasets to bootstrap pediatric classification tasks via few-shot or zero-shot adaptation. In one such study, a model pretrained on adult chest X-rays using a multimodal contrastive loss was evaluated on pediatric pneumonia detection using only 10 labeled examples [47]. By embedding task prompts such as “bilateral infiltrates in pediatric chest X-ray,” the model could retrieve semantically similar images and achieve AUROC scores above 0.85, significantly outperforming shallow CNN baselines. Furthermore, combining text prompts with uncertainty-aware loss functions helped identify ambiguous or atypical presentations, a crucial feature in rare disease contexts.

4.4 Histopathology and Whole-Slide Image (WSI) Analysis

Digital pathology presents unique challenges for domain adaptation and generalization due to the enormous size of whole-slide images (WSIs), variability in staining protocols, and strong inter-scanner differences. Color normalization, tile-level augmentation, and adversarial feature alignment have been traditional remedies for domain shift, but these approaches often lack robustness in large-scale settings [48]. Foundation models pretrained on patch-level pathology tiles, such as PathCLIP and BioViL-Histo, have emerged as powerful alternatives. These models are trained using millions of paired histology tiles and diagnostic text descriptions (e.g., pathology reports or biopsy summaries) [49]. In one case study, PathCLIP was evaluated on breast cancer grading across three international datasets with different staining standards [50]. The model achieved consistent performance across domains using zero-shot inference, and performance improved further with adapter tuning using a small labeled support set from each target domain. The model also

exhibited calibrated confidence scores and interpretable attention maps, facilitating clinical acceptance [51].

4.5 Multi-Center Federated Learning in Radiology

Federated learning (FL) is increasingly being adopted in multi-institutional collaborations where data privacy constraints prevent centralized training [52]. However, heterogeneity across client datasets creates significant domain shifts that degrade model convergence and generalization [53]. Federated domain adaptation and generalization aim to address this by incorporating domain-aware updates, personalization layers, or model ensembling. Recent FL frameworks have integrated foundation models as the shared initialization point across clients [54]. For instance, a foundation model pretrained on large-scale chest X-ray datasets was distributed across five hospitals for federated fine-tuning on pneumonia and COVID-19 detection. Compared to training from scratch, the foundation-initialized model exhibited faster convergence and better performance on out-of-network test sets. Additionally, fine-tuning only lightweight adapters or batch norm layers allowed each client to personalize their model while preserving global consistency. This case highlights the role of foundation models in harmonizing federated updates across non-IID medical datasets.

4.6 Radiology Report Generation and Vision-Language Reasoning

The capability of foundation models to jointly reason over images and text has enabled applications beyond classification or segmentation, including report generation, retrieval, and visual grounding [55]. Models like BioGPT, GatorTron, and BioViL have been combined with vision backbones to generate structured radiology reports, retrieve relevant historical studies, or answer clinical queries [56]. In one notable deployment, a vision-language foundation model trained on image-report pairs from a national radiology archive was fine-tuned to generate summaries for chest CT exams. During evaluation on three external datasets with differing reporting styles and imaging protocols, the model maintained high BLEU and ROUGE scores while preserving semantic accuracy in key clinical findings [57]. These results underscore the importance of cross-modal pretraining and prompt engineering in achieving robust generalization in real-world NLP+vision medical tasks [58].

4.7 Empirical Lessons and Limitations

Across these applications, several empirical trends emerge:

- Foundation models offer significant improvements in generalization compared to traditional DA/DG methods, especially when pretrained on diverse and semantically rich data.
- Prompt tuning and text conditioning are powerful tools for zero-shot inference, but prompt engineering remains more art than science, and models are sensitive to prompt phrasing [59].
- Adapter-based fine-tuning strikes a balance between performance and computational efficiency, particularly in low-resource or federated settings [60].
- While synthetic data generation and pixel-level translation are promising for domain adaptation, their clinical reliability is still debated, necessitating careful validation.
- Most current benchmarks remain small-scale or biased toward specific tasks; larger and more diverse evaluation protocols are essential to assess true generalization [61].

In summary, the application of foundation models for domain adaptation and generalization in medical imaging has moved from theoretical potential to practical impact. From radiology to pathology and beyond, these models are demonstrating the ability to overcome longstanding limitations in data heterogeneity, scarcity, and distributional shift. In the subsequent sections, we will critically analyze the open challenges, ethical implications, and future research directions that must be addressed to ensure safe, equitable, and scalable deployment of these technologies in clinical practice [62].

5 Challenges, Limitations, and Ethical Considerations

Despite the growing enthusiasm and demonstrated successes in using foundation models for domain adaptation (DA) and domain generalization (DG) in medical imaging, a number of critical challenges remain. These span across technical, operational, and ethical domains, posing serious obstacles to large-scale, trustworthy deployment [63]. In this section, we delineate the key limitations and discuss the broader implications of relying on foundation models in sensitive medical contexts.

5.1 Data Bias and Representation Inequality

A foundational challenge in domain generalization stems from the bias inherent in the pretraining datasets [64]. Most foundation models in medical imaging are trained on large but heterogeneous data collections, which are often dominated by a few well-resourced institutions, high-resource regions, and prevalent disease types [65]. This leads to skewed representations that may not generalize well to underrepresented patient populations, imaging modalities, or clinical settings. For instance, a model trained predominantly on adult chest X-rays from U.S. [66, 67]. Hospitals may underperform on pediatric cases or data from low- and middle-income countries (LMICs). This exacerbates healthcare disparities and could propagate existing inequalities if such models are deployed without proper auditing. Moreover, public datasets are frequently curated for research convenience rather than clinical realism, introducing further bias in acquisition protocols, label quality, and disease prevalence [68].

5.2 Domain Shift Beyond Visual Distribution

While DA and DG methods focus primarily on visual domain shifts (e.g., scanner types, imaging artifacts), real-world clinical deployment encounters far more complex domain shifts [69]. These include shifts in diagnostic practices, annotation standards, population health profiles, and even the clinical meaning of certain patterns due to changing guidelines or disease definitions. Foundation models pre-trained on historical datasets may therefore be ill-equipped to handle these evolving clinical semantics. In addition, non-visual metadata—such as demographics, comorbidities, and treatment histories—plays a critical role in diagnosis but is often missing or inconsistently structured in the training data. This limitation constrains the ability of foundation models to perform holistic reasoning or causal inference across domains, even when their visual representations appear robust.

5.3 Interpretability and Trustworthiness

Interpretability remains a major barrier to the adoption of foundation models in clinical practice [70]. While many DA and DG methods enhance performance, they often obscure model decision-making through complex architectures or adversarial losses. Foundation models, particularly those based on large-scale self-supervision or vision-language learning, further complicate interpretability due to their opaque and distributed representations. Although some methods, such as attention heatmaps or saliency-based techniques, are used to visualize model focus areas, these approaches are not always reliable and may fail under domain shift [71]. For example, a model trained on natural images may misattribute saliency

to irrelevant features in histopathological images, undermining clinical trust. The black-box nature of foundation models thus necessitates rigorous interpretability tools and domain-specific validation protocols before deployment [72].

5.4 Prompt Sensitivity and Reliability in Zero-Shot Settings

One of the strengths of foundation models is their ability to perform zero-shot inference using natural language prompts or image embeddings. However, prompt engineering introduces new sources of variability and unreliability [73]. Small changes in phrasing, syntax, or terminology can result in significant fluctuations in model outputs, especially when prompts are not grounded in clinical standards. In radiology, for instance, changing the prompt from “consolidation” to “lung opacity” can yield different predictions, even though both terms refer to similar phenomena. Moreover, foundation models may exhibit bias toward frequently seen terms during pretraining, neglecting rare or subtle conditions [74]. Ensuring robustness across prompts and establishing best practices for prompt formulation are ongoing challenges that require both algorithmic innovation and clinical collaboration.

5.5 Computational and Infrastructural Barriers

Training and deploying foundation models—especially in the context of DA and DG—entails significant computational resources, data storage, and engineering expertise. This limits accessibility for smaller institutions, research labs, and LMICs, where computational infrastructure may be lacking. Moreover, the inference cost of large models may be prohibitive in real-time or edge deployment scenarios, such as portable ultrasound or point-of-care diagnostics. Although techniques such as adapter tuning, quantization, and knowledge distillation can alleviate some of these burdens, there remains a need for more efficient and scalable frameworks that democratize access to foundation model capabilities [75]. Without such solutions, the benefits of foundation model generalization may remain concentrated among a few elite institutions.

5.6 Ethical Risks and Legal Uncertainty

The use of foundation models in clinical environments raises a host of ethical and legal concerns. These include:

- **Privacy and data governance:** Large-scale pretraining often involves data aggregation across multiple sources, raising questions about patient consent, de-identification robustness, and data reuse.

- **Liability and accountability:** In the event of clinical error, it is unclear how responsibility is apportioned between developers, clinicians, and institutions, particularly when the model was adapted or fine-tuned post-deployment.
- **Bias amplification:** Foundation models may inadvertently reinforce diagnostic disparities by underperforming on minority populations or rare disease subtypes, especially when not explicitly evaluated for fairness.
- **Dual use and misuse:** General-purpose foundation models can be repurposed for non-clinical or even harmful applications, such as unauthorized surveillance or insurance risk modeling, necessitating careful access controls.

Existing regulatory frameworks, including FDA guidelines and CE certification pathways, are not yet fully equipped to handle the fluid and evolving nature of foundation models, especially those capable of continual adaptation across domains [76]. Establishing standards for transparency, auditing, and post-market surveillance is imperative to ensure safe and equitable deployment [77].

5.7 Benchmarking and Evaluation Gaps

Current evaluation practices for DA and DG in medical imaging often rely on narrow benchmarks, limited metrics, and synthetic domain shifts that fail to capture real-world complexity [78]. For instance, testing on a held-out institution or modality may not reflect the model’s performance under subtle but clinically significant variations, such as protocol changes or rare comorbidities. Moreover, many studies report performance gains in terms of accuracy or Dice score but neglect to assess calibration, uncertainty, sample efficiency, or fairness across subpopulations. This hinders meaningful comparison between methods and obscures the practical implications of performance differences. The field would benefit greatly from the development of large-scale, multi-domain, community-curated benchmarks designed specifically for DA and DG evaluation, along with richer metrics aligned with clinical utility [79].

In summary, while foundation models present a promising frontier for robust and generalizable medical imaging AI, their practical realization is fraught with challenges. From data and infrastructure inequities to interpretability concerns and ethical ambiguity, there exists a wide chasm between experimental success and clinical adoption [80]. These issues must be addressed not only through technical innovation but also through interdisciplinary collaboration involving clinicians, ethicists, regulators, and patient communities [81]. In the following section, we

discuss promising future directions aimed at bridging this gap and unlocking the full potential of foundation models for domain-robust medical imaging.

6 Future Directions and Open Problems

The intersection of foundation models with domain adaptation (DA) and domain generalization (DG) in medical imaging represents a rapidly evolving and profoundly interdisciplinary research landscape. As the field matures, it is becoming increasingly clear that realizing the full potential of these models requires rethinking existing paradigms across model design, data governance, deployment infrastructure, and human-AI interaction. This section identifies emerging trends, articulates unresolved challenges, and outlines promising research directions that will shape the next generation of generalizable and clinically reliable medical AI systems [82].

6.1 Unified Frameworks for Multi-Level Generalization

While current approaches to domain adaptation and generalization are typically designed around specific tasks (e.g., classification, segmentation) or domain shifts (e.g., modality, institution), there is growing interest in developing unified frameworks that can generalize across multiple axes simultaneously [83]. Such systems would need to account for a broad spectrum of shifts—acquisition artifacts, demographic variation, anatomical diversity, and even temporal drift in clinical practices [84]. One promising avenue involves hierarchical or modular architectures that combine global foundation model backbones with task- or domain-specific adapters, enabling fine-grained control over knowledge transfer [85]. Another possibility is to leverage meta-learning and continual learning paradigms, allowing models to accumulate and refine knowledge over time as they encounter new environments and tasks [86]. Future research should explore how to design these systems to be not only flexible and scalable but also robust to catastrophic forgetting and overfitting in heterogeneous clinical settings.

6.2 Causal and Counterfactual Approaches to Domain Robustness

Most domain adaptation and generalization strategies are based on statistical alignment or representation learning, which may fail under spurious correlations or out-of-distribution data [87]. Integrating causal reasoning into foundation models offers a promising alternative, enabling models to disentangle invariant causal factors from superficial correlations. Future work could explore how to embed causal

graphs, structural equation models, or counterfactual reasoning mechanisms into the architecture of foundation models. Such approaches may enhance the models’ ability to generalize by identifying stable relationships across domains, particularly in high-stakes applications such as disease diagnosis, prognosis, and treatment planning [88]. Importantly, this also opens the door for integrating mechanistic knowledge from clinical guidelines and pathophysiological models, thereby increasing trust and transparency [89].

6.3 Domain-Aware Pretraining and Curriculum Learning

Current pretraining paradigms for foundation models are largely domain-agnostic, emphasizing scale over specificity. However, medical imaging presents unique challenges and opportunities for domain-aware pretraining. This includes leveraging structured knowledge (e.g., ontologies like SNOMED-CT or RadLex), modality-specific cues (e.g., CT windowing, MRI sequences), and progressive curricula that reflect real-world diagnostic workflows [90]. Curriculum learning—where models are exposed to increasingly complex or diverse domains over time—can be a valuable tool for simulating clinical learning processes. Future research should investigate optimal strategies for organizing and sequencing domain exposure during pretraining, as well as mechanisms for dynamic curriculum adjustment based on model uncertainty or task complexity [91]. These methods may yield more robust and sample-efficient models, especially in settings with limited labeled data [92].

6.4 Self-Supervised and Semi-Supervised Learning for Rare and Low-Resource Domains

Many of the most critical applications of generalizable medical AI lie in low-resource settings, rare disease identification, and underserved populations. In these cases, collecting large labeled datasets is either infeasible or ethically challenging. Foundation models trained with self-supervised or semi-supervised objectives offer a compelling solution. Future work could focus on developing targeted pretext tasks and data augmentations that are optimized for rare pathological patterns, pediatric anatomy, or low-quality imaging modalities. Additionally, research is needed to evaluate how well foundation models trained on high-resource domains can transfer to these low-resource settings without amplifying biases or degrading performance [93]. Exploring efficient fine-tuning strategies, such as LoRA (Low-Rank Adaptation) or prompt-based regularization, will be critical for scaling these models in data-sparse environments.

6.5 Benchmarking, Simulation, and Real-World Evaluation

A significant limitation of current research is the lack of comprehensive and clinically grounded benchmarks for DA and DG. Most existing evaluations rely on a narrow set of public datasets and often simulate domain shift in artificial ways [94]. There is an urgent need for community-led initiatives to establish large-scale, multi-institutional, multi-modality benchmarks that reflect real-world diversity in patients, scanners, protocols, and pathologies [95]. Simulated environments and digital twins may also serve as valuable testbeds for evaluating generalization under controlled yet clinically plausible conditions [96]. For example, generative models could create synthetic cohorts with specific anatomical, demographic, or disease characteristics to stress-test foundation models. Complementary real-world evaluation—via prospective clinical studies, regulatory sandboxes, and shadow deployments—will be essential to assess utility, safety, and trustworthiness before widespread adoption.

6.6 Human-AI Collaboration and Interactive Adaptation

Most domain adaptation methods assume a passive model that is trained once and deployed as-is. However, clinical workflows are inherently interactive, and domain shifts may emerge gradually over time. Future systems should embrace this interactivity by supporting human-in-the-loop learning, model editing, and feedback-driven adaptation. This vision entails new user interfaces and training paradigms that allow clinicians to guide model behavior through example selection, natural language feedback, or visual annotation. Research on interactive adaptation—where models can dynamically refine their representations based on human input or institutional drift—will be pivotal for creating resilient and adaptive AI assistants [97]. These systems must also be transparent about their uncertainties and limitations, allowing clinicians to intervene when generalization fails.

6.7 Ethics, Fairness, and Participatory Model Development

The development of generalizable AI systems must be accompanied by rigorous attention to ethics, equity, and inclusivity. Future research should prioritize participatory approaches to model design that involve patients, clinicians, and community stakeholders from the outset. This includes mechanisms for algorithmic auditing, bias mitigation, and post-deployment monitoring, as well as frameworks for informed consent and data ownership [98]. Moreover, regulatory frameworks

need to evolve to accommodate foundation models, especially those capable of continual learning or domain-specific adaptation. International collaboration among researchers, regulators, and policy-makers will be critical to harmonize standards and ensure that technological advances do not exacerbate global health inequities [99].

In summary, while significant progress has been made, the journey toward truly domain-robust and generalizable medical imaging AI is only beginning. Foundation models offer a powerful substrate upon which more adaptive, interpretable, and equitable systems can be built, but realizing this vision demands interdisciplinary innovation, community-wide collaboration, and unwavering commitment to ethical integrity. By addressing these open problems, the field can move closer to creating medical AI systems that are not only technically excellent but also clinically meaningful and globally inclusive.

7 Conclusion

The convergence of foundation models and domain adaptation/generalization techniques represents a transformative moment for medical imaging AI. As diagnostic imaging continues to play a central role in clinical decision-making, the ability of AI systems to operate reliably across diverse patient populations, imaging devices, and healthcare environments becomes increasingly vital. Foundation models, with their scalability, multimodal capacities, and potential for zero-shot inference, offer an unprecedented opportunity to build robust and adaptable systems that transcend conventional domain boundaries.

In this survey, we have examined the fundamental motivations for addressing domain shift in medical imaging, reviewed the spectrum of strategies for domain adaptation and generalization, and critically assessed the role of foundation models in enabling these capabilities. We highlighted the technical mechanisms that underpin recent advances—ranging from self-supervised learning to adapter-based fine-tuning—and considered their limitations in terms of interpretability, fairness, and clinical readiness. Importantly, we emphasized that generalization in medical imaging is not solely a matter of algorithmic sophistication but also a complex socio-technical challenge that requires attention to data equity, regulatory alignment, and human-centered design.

Our exploration of challenges revealed that significant barriers remain, including biased pretraining data, prompt sensitivity, high computational costs, and the lack of standardized benchmarks. These issues must be addressed with deliberate and interdisciplinary effort. Meanwhile, the research landscape continues to evolve, pointing toward promising directions such as causal generalization, curriculum-

aware pretraining, simulation-driven evaluation, and interactive human-AI adaptation. These emerging paradigms hold the potential to transform foundation models from static assets into dynamic collaborators in clinical workflows.

Ultimately, the goal of domain-robust foundation models in medical imaging is not merely to improve accuracy on benchmark datasets, but to deliver tangible, equitable benefits to patients across geographies, modalities, and healthcare systems. This requires not only algorithmic innovation but also a reimagining of how models are trained, evaluated, deployed, and governed. As the field moves forward, collaboration between machine learning researchers, radiologists, ethicists, policy-makers, and patient communities will be essential. Together, they can chart a path toward AI systems that are not only powerful and generalizable but also trustworthy, inclusive, and responsive to the diverse realities of global health.

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